

Retail Business Performance & Profitability Analysis

1. Abstract

This project analyzes the Superstore retail dataset using Python, SQL, and Power BI to evaluate sales, profit, category performance, regional performance, sub-category profitability, and discount impact. The analysis identifies profitable business areas and loss-making segments and provides practical recommendations to improve profitability.

2. Introduction and Dataset Overview

The objective of this project is to understand overall retail business performance and identify opportunities to improve profit. The dataset contains 10,194 sales records and 21 columns, including order details, customer information, product categories, sales, quantity, discount, and profit. The dataset was inspected for data quality and missing values before analysis.

3. Tools Used and Methodology

Tools	Excel, Python, Pandas, Matplotlib, MySQL, and Power BI
Methodology	Inspected and prepared the dataset; performed SQL aggregation and profitability analysis; created Python charts; developed an intera

4. Overall Business Performance

The business generated total sales of \$2,326,534.35 and total profit of \$292,296.81. A total of 38,654 items were sold, and the average discount offered was 15.54%. These results show that the business is profitable overall, although profitability varies across categories, regions, products, and discount levels.

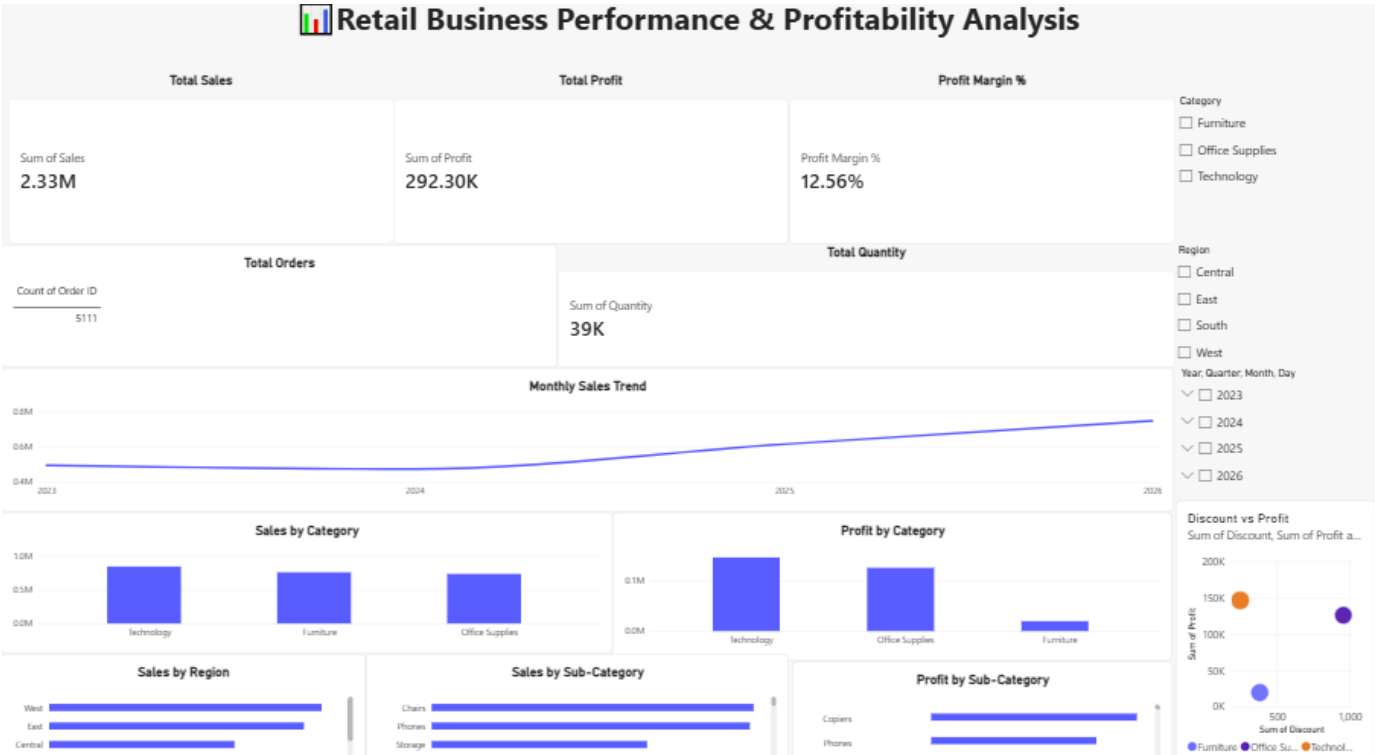
5. Category and Sub-Category Performance

Technology was the strongest-performing category, generating \$839,893.28 in sales and \$146,543.38 in profit. Office Supplies generated \$731,893.31 in sales and \$126,023.44 in profit. Furniture generated \$754,747.76 in sales but only \$19,730.00 in profit, indicating weaker profitability despite substantial sales.

Copiers generated the highest sub-category profit at \$56,093.94. Other strong sub-categories included Phones, Accessories, Paper, and Binders. The loss-making sub-categories were Tables (-\$17,753.21), Bookcases (-\$3,632.07), and Supplies (-\$1,171.39). Tables are the biggest concern because they generated \$208,020.18 in sales but still resulted in a large loss.

6. Regional and Discount Analysis

The West region performed best, with \$739,813.61 in sales and \$110,798.82 in profit. The East region earned \$94,883.26 profit, while the Central region earned \$39,865.31 profit despite having higher sales than the South region. Discounts up to 20% were profitable overall, whereas discounts of 30% and above resulted in losses. The largest loss occurred at a 70% discount, with total profit of -\$40,300.57.



7. Key Findings and Recommendations

- Reduce discounts above 20%, especially discounts between 30% and 80%.
- Review pricing, discount policies, and costs for Tables, Bookcases, and Supplies.
- Investigate loss-making states such as Texas, Ohio, Pennsylvania, Illinois, and North Carolina.
- Focus marketing efforts on profitable Technology products such as Copiers, Phones, and Accessories.
- Study successful strategies used in the West region and apply suitable practices to lower-performing regions.

8. Conclusion

The business is profitable overall, with strong performance in Technology and the West region. However, high discounts, loss-making sub-categories, and certain states reduce profitability. Better discount control, improved pricing decisions, and focused attention on weak product and regional segments can help increase overall profit.